

Statement of Purpose

My research focuses on **design-based causal inference** for observational studies and on **statistical learning under distribution shift**. More broadly, I am interested in how inference should proceed when the mechanism that generated the data is only partially known: when treatment may depend on unmeasured covariates, when a deployed model meets data from a shifted distribution, or when a variable was produced by another model rather than measured. My research has led to a **co-first-author** working paper on sensitivity analysis for matched observational studies [1], a co-authored manuscript submitted to *Science* [2], and a **first-author** manuscript on audio-visual test-time adaptation under submission [3], together with co-authored papers under major revision at *IEEE TPAMI* [4] and *ACM TKDD* [5] and one submitted to AAAI 2027 [6].

I completed my B.Sc. in Mathematics through the 2+2 programme of Xi'an Jiaotong-Liverpool University and the University of Liverpool, graduating with **First Class Honours** and ranked in the **top 1%** of my major, with marks of 99 in Further Methods of Applied Mathematics, 95 in Asymptotic Methods for Differential Equations, 93 in Waves and Mathematical Modelling, and 92 in Statistics and Probability. Since September 2025 I have been a master's student in Computational and Applied Mathematics at the University of Chicago, where my coursework includes Advanced Topics in Causal Inference, High-Dimensional Probability, Random Matrix Theory, Inverse Problems and Data Assimilation, Nonlinear Optimization, and Foundations of Machine Learning and AI. Before Chicago, as a machine learning research intern at iFLYTEK, I trained Transformer and Conformer models on in-house text and speech data and improved BLEU from 27.6 to 29.9 and word error rate from 10.3% to 8.9%. The models did well on the metrics, but I could say nothing about when they would stop working and nothing about what would have happened under a different intervention. That gap, between fitting a model and being able to defend a claim, is what drew me to statistics. In the following, I outline the research directions I have explored and aim to build upon during my PhD.

Covariate-Adaptive Sensitivity Analysis for Matched Observational Studies: Matched observational studies are usually analyzed as if treatment had been assigned uniformly within each matched set, yet inexact matching leaves systematic propensity-score discrepancies that can invalidate the naive randomization test [7]. Covariate-adaptive randomization inference [8] replaces the uniform law with probabilities derived from fitted propensity odds, and studentized sensitivity analysis [9] tests Neyman's weak null on the average effect under Rosenbaum's hidden-bias model [10]. No existing method does both. Together with Pengyun Wang, under the guidance of Prof. Xinran Li, I developed a **covariate-adaptive studentized sensitivity analysis** for Neyman's weak null. A unit-level hidden-bias bound Γ_u induces, after conditioning on one treated unit per pair, a pair-level parameter $\Lambda = \Gamma_u^2$ and a pair-specific interval for the conditional assignment probability, anchored at the fitted propensity odds rather than at one half. I proposed an **asymmetric Horvitz–Thompson lower envelope** over this interval and proved that it is mean-controlled and pointwise sharp within a zero-normalized observed-cell class, that it reduces to the known-propensity Horvitz–Thompson statistic when $\Lambda = 1$, and that studentizing it yields asymptotically valid tests, with a randomization-calibrated critical value and an honest inflated-endpoint version for estimated propensities. I then extended the construction to **full matching** [11] in closed form: a matched set with more than two units has a probability region rather than an interval, but a level-free difference contrast depends on the assignment only through one realized coordinate, whose marginal range is again a scalar interval, so the entire studentized theory carries over. The within-set mean turns out to be one member of a family of balanced matrix-weighted contrasts that share every guarantee, and its propensity-anchored local members recover substantial power in large heterogeneous sets.

In the right heart catheterization study, where 1,386 matched pairs retain measurable propensity discrepancy, the covariate-adaptive analysis reports a sensitivity value of $\Lambda = 1.40$ against 1.30 for the uniform analysis, while a symmetric envelope centered at one half fails to reject at all; accounting for residual discrepancy materially changes the reported robustness of a conclusion. The manuscript [1] is a working paper with equal contribution from Pengyun Wang and me.

This project taught me to treat the assignment mechanism as the object of inference and to trace every claim back to a distribution over assignments that I am prepared to defend. It also gave me the question that organizes my research proposal. A sensitivity model bounds how far the true assignment law may sit from an anchor estimated from data, and reports a guarantee that holds over the whole band. The same structure, an anchor, a bounded departure, and a guarantee over the band, is what machine learning under distribution shift needs and mostly lacks, which is why I pursued the two projects below in parallel.

Test-Time Adaptation under Multimodal Distribution Shift: Test-time adaptation (TTA) updates a pretrained model online from unlabeled target batches, typically by entropy minimization [12, 13]. In audio-visual streams, corruptions hit the two modalities asymmetrically, and a corrupted modality enters the optimizer twice: it shapes the prediction and then returns as its own pseudo-label, so errors accumulate. With Prof. Xiao Luo at the University of Wisconsin–Madison, I developed **HEART**, which treats a confident pseudo-label as a hypothesis to be checked rather than as a training target. Two half-routers, each combining one original and one adapted representation, test whether complementary audio-visual evidence still supports the anchor label; the label is used only when both routes agree, and a batch gate built from support coverage and support contrast weights the pseudo-labeling objective. On the VGGSound-2C benchmark of AVRobustBench [14] (15 paired corruptions at severity 5), HEART raises mean accuracy from 41.35% for the source model to **48.90%**, above the strongest prior method AV2C at 47.31% [15], while TENT falls to 34.89%. An earlier version of this work, built on binary modality accept/reject gating, entropy-adaptive token dropout, and class-conditional thresholds, also showed that entropy-minimization baselines are implementation-sensitive: enabling EATA’s official redundancy filter moves it from 9.2% to 31.5% on a severe-noise subset, a fragility that prior reporting had hidden. The first-author manuscript [3] is under submission. In the same period I contributed to a graph pooling benchmark that evaluates robustness under noise and out-of-distribution shift [5] and to a multimodal fusion framework for sentiment analysis [4], and I served as a reviewer for the Forecast@ICML 2026 workshop.

Scientific World Models for Research Agents: LLM agents now run machine learning experiments on their own [16], but they plan poorly because they cannot predict what an experiment will return before running it. With Prof. Zhuang Liu at Princeton, I am building a **scientific world model** that maps a research state and a candidate action to a predicted outcome. On research trees of up to 1,240 executed transitions, state-aware predictors reduced planner regret by about **80%** in held-out replay, but leave-one-action-family-out evaluation dropped pairwise ranking accuracy to 0.34, exposing an out-of-distribution failure. I designed mechanism-aware predictors with structured intervention representations and hierarchical backoff, which raised out-of-distribution pairwise accuracy to 0.59 within action families and 0.69 across the mixed frontier with a 0.79 regret reduction, and then a test-time selective router that chooses among frozen predictors from shift diagnostics. The most useful result was statistical rather than architectural. A replicate audit that re-ran contested action pairs under multiple training seeds showed that some families are seed noise while others depend on the parent state, and a diagnosis-aware policy that ranks only when the evidence is stable achieved **100% accuracy** on the 47% of pairs it chose to rank, against 71% for a policy that always ranks. A world model must know when to abstain, and execution evidence is

a distribution over outcomes rather than a label. I now apply the same discipline to the agent's written output, building a checker that scores generated research reports against their own logs and treats a fabricated number as disqualifying.

Modeling the Dynamics of Scientific Attention: At the Knowledge Lab with Prof. James Evans, I led the mathematical modeling for a study of how AI breakthroughs concentrate scientific attention and erode the intellectual diversity their successors require, drawing on 7.18 million papers, 73 thousand peer reviews, and 3.7 million scientists over 36 years [2]. Because the peer-review policy of interest arrived together with large language models, causal identification from observational data was structurally out of reach, so with Honglin Bao I designed and implemented an agent-based model of reciprocal peer review in which reviewer assignment scales with submission load and taste-driven scoring concentrates accepted topics; across parameter sweeps and 25 seeds per cell, stronger reciprocity produced faster collapse of intellectual diversity. I also worked on the logistic-growth model with post-saturation decay that describes the lifecycle of active ideas within each paradigm cycle. Working alongside sociologists showed me how often the variables in modern empirical work, topics, novelty, stance, or severity extracted from text, are produced by models rather than measured, and how rarely that error is carried into the inference. My research proposal takes this up directly: it develops design-based sensitivity analysis for matched studies whose data depart from the ideal in three ways at once, through hidden bias, through population shift, and through model-generated measurements.

Why a PhD at NUS: I am applying to the PhD in Statistics and Data Science at NUS because the department combines the two things my proposal needs, rigorous statistical theory and serious engagement with modern data, in one place. My preferred supervisor is **Prof. Dongming Huang**, whose work on conditional inference via exchangeable sampling and on catalytic priors, which use synthetic data to stabilize estimation without giving up validity, provides exactly the tools I need when model-generated data enter a matched analysis as auxiliary information; I had the opportunity to discuss my research with Prof. Huang in September 2026, and the conversation confirmed how closely our interests overlap. **Prof. Yao Zhang**'s work on randomization tests and on average-case sensitivity analysis for unmeasured confounding bears directly on the design-sensitivity and weak-null questions in my proposal, and the same group's work on conformal prediction on the abstention rules I want to make rigorous. **Prof. Doudou Zhou**'s work on statistical evaluation under covariate shift, on inference with surrogate-derived models, and on evaluating language models with cheap auxiliary signals is relevant to the population-shift and model-generated-measurement parts of the proposal, and clinical text is one of my two planned application areas. I would also value the department's broader causal inference community, including **Prof. Baoluo Sun** and **Prof. Jinzhou Li**. Over the past year I have shown that I can carry several projects at once, work productively with groups at institutions other than my own, and take a result from conjecture to proof to application. I am applying for a PhD because the questions I now care about need more time and more than one paper, and because I want an academic career in which I keep working on them and teach others to. I would be glad to begin at NUS in August 2027.

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